Yiqi Wang

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Department of Physics, Harvard University,
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Education

- **Yale University** *Aug 2017 – Aug 2023*
  Ph. D. | *Applied Physics* New Haven, CT
  - Dissertation title: *Manipulating and Measuring States of a Superfluid Optomechanical Resonator in the Quantum Regime*
  - Doctoral thesis with distinction
  - Advisor: Jack G. E. Harris

- **Fudan University** *Sep 2013 – Jun 2017*
  B. S. | *Physics* Shanghai, China
  - Shanghai Outstanding Graduate
  - Exchange student, The Chinese University of Hong Kong (Fall 2015)

Research Experience

- **Harvard University** *Aug 2023 – Present*
  HQI Postdoctoral Fellow Cambridge, MA
  - PI: Mikhail D. Lukin
    - Interfacing a mechanical oscillator with a single electron spin.
    - Levitodynamics of a micron-scale levitated magnet.

- **Yale University** *Aug 2017 – Aug 2023*
  Research Assistant New Haven, CT
  - PI: Jack G. E. Harris
    - Quantum control and readout of phonon states in a superfluid-helium-filled Fabry–Pérot fiber cavity.
    - Levitating superfluid helium droplets in vacuum for quantum optomechanics.

Research Interests

Quantum optomechanics, Quantum sensing, Quantum computing, Solid state qubit, Hybrid quantum system

Publications

J=Journal, T=Thesis, S=In Submission, P=In Preparation, *:Equal Contribution

[P.1] T. Madhavan*, Y. Wang*, A. NewRingeisen, J. D. Schaefer, F. Fung, M. D. Lukin.
Micrometer-Scale Magnetometry based on a Levitated-Magnet: From $\mathrm{pT}/\sqrt{\mathrm{Hz}}$ sensitivity toward Quantum-Enhanced Sensing.

[P.2] Y. S. S. Patil,... Y. Wang, *et al.*, J. G. E. Harris.
Thermal-limited Motion of a Levitated Superfluid Optomechanical System.

[P.3] V. Wachter, Z. Wei, Y. Wang, T. Madhavan, J. D. Schaefer, A. NewRingeisen, M. D. Lukin, B. A. Stickler.
Rotational Dynamics of Meissner-levitated Micromagnets.

[P.4] Y. Wang, *et al,* J. G. E. Harris.
Testing the Linearity of Quantum Evolution with Macroscopic Motion.

[P.5] Y. Wang*, Y. S. S. Patil*, *et al,* J. G. E. Harris.
Entangling Light and Sound in Superfluid Helium.

[S.1] Y. Wang*, T. Madhavan*, V. Wachter*, J. D. Schaefer, A. NewRingeisen, Z. Wei, F. Fung, B. A. Stickler, M. D. Lukin.
Nonlinear Dynamics and Mechanical Frequency Combs with a Meissner-levitated Micromagnet.
arXiv: 2609.11668 (2026).
manuscript submitted to *Nature Physics*.

[S.2] Y. Wang*, T. Madhavan*, J. D. Schaefer, A. NewRingeisen, F. Fung, M. D. Lukin.
Exploring Dynamics of Individual Vortices in a Superconductor via a Levitated Magnetic Transducer.
arXiv: 2606.27297 (2026).
manuscript submitted to *Physical Review X*.

[S.3] J. Ma, Y. S. S. Patil, J. Yu, Y. Wang, J. G. E. Harris.
A Proposal for Detecting the Spin of a Single Electron in Superfluid Helium.
arXiv: 2308.07174 (2023).

[T.1] Y. Wang.
Manipulating and Measuring States of a Superfluid Optomechanical Resonator in the Quantum Regime.
Yale doctoral thesis (2023).

[J.1] C. D. Brown, Y. Wang, M. Namazi, G. I. Harris, M. T. Uysal, J. G. E. Harris.
Superfluid Helium Drops Levitated in High Vacuum.
Physical Review Letters, 130, 216001 (2023).
Press: *New Scientist*.

[J.2] Y. S. S. Patil, J. Yu, S. Frazier, Y. Wang, K. Johnson, J. Fox, J. Reichel, J. G. E. Harris.
Measuring High-Order Phonon Correlations in an Optomechanical Resonator.
Physical Review Letters, 128, 183601 (2022).

Talks And Presentations

O=Oral, P=Poster

[O.19]	Annual Meeting of DAMOP, *Providence, RI*	*Jun 2026*
[O.18]	Quantum Phononic MURI Grant Review, *Arlington, VA*	*Apr 2026*
[O.17]	APS March Meeting, *Denver, CO*	*Mar 2026*
[P.11]	Center for Ultracold Atoms Winter Conference, *Plymouth, NH*	*Jan 2026*
[P.10]	HQI-RIKEN Quantum Workshop, *Cambridge, MA*	*Oct 2025*
[O.16]	Quantum Phononic MURI Grant Review, *Arlington, VA*	*May 2025*
[O.15]	APS March Meeting, *Anaheim, CA*	*Mar 2025*
[P.9]	Center for Ultracold Atoms Winter Conference, *Plymouth, NH*	*Jan 2025*
[P.8]	Spin-Mechanics Workshop, *Portland, OR*	*Aug 2024*
[O.14]	HKUST Physics Seminar, *Hong Kong (Virtual)*	*Aug 2024*
[O.13]	Quantum Information and Technology Conference, *Shanxi (Virtual)*	*Jul 2024*
[P.7]	Quantum Noir, *Cambridge, MA*	*Jun 2024*
[O.12]	Quantum Phononic MURI Grant Review, *Arlington, VA*	*May 2024*
[P.6]	Gordon Research Conference, *Ventura, CA*	*Mar 2024*
[P.5]	Center for Ultracold Atoms Winter Conference, *Plymouth, NH*	*Jan 2024*
[O.11]	Quantum Phononic MURI Grant Kick-off, *New Haven, CT*	*Nov 2023*
[O.10]	APS March Meeting, *Las Vegas, NV*	*Mar 2023*
[O.9]	McGill Special CPM Seminar, *Montreal, Canada*	*Jan 2023*
[O.8]	Université de Sherbrooke Séminaire, *Sherbrooke, Canada*	*Jan 2023*
[O.7]	JILA Science Seminar, *Boulder, CO*	*Jan 2023*
[P.4]	Winter colloquium on the Physics of Quantum Electronics (PQE), *Snowbird, UT*	*Jan 2023*
[O.6]	Harvard Quantum Initiative Quantum Fest, *Cambridge, MA*	*Dec 2022*
[O.5]	Yale Science in the News, *New Haven, CT*	*Oct 2022*
[P.3]	International Conference on Atomic Physics (ICAP), *Toronto, Canada*	*Jul 2022*
[P.2]	Gordon Research Conference, *Ventura, CA*	*Jun 2022*
[P.1]	Optica Quantum 2.0 Conference and Exhibitions, *Boston, MA*	*Jun 2022*
[O.4]	APS March Meeting, *Chicago, IL*	*Mar 2022*
[O.3]	NONGAUSS Workshop, Virtual	*May 2021*
[O.2]	APS March Meeting, Virtual	*Mar 2021*
[O.1]	Yale Quantum Institute Talk, *New Haven, CT*	*Apr 2019*

Funding and Proposal Experience

- **Quantum Phononics Multidisciplinary University Research Initiative (MURI)** *2023 – Present*
  Contributing researcher (Corresponding PI: M. D. Lukin; Lead PI: Konrad Lehnert)
  - Drafted three annual technical reports.
  - Presented project progress at four annual program reviews (2023–2026).
- **"Quantum Co-design of Materials and Mechanisms for Electro-acousto-optical Transduction"** *Submitted 2026*
  Contributor (Corresponding PI: M. D. Lukin; Lead PI: Jennifer Hoffman)
  - Wrote the quantum-sensing pedagogical box and contributed to the thrust on system integration and transduction-enabled quantum applications.

Service

- **APS March Meeting** *2023, 2026*
  Session Chair
- **Yale Physics Department Graduate Open House** *2022, 2023*
  Graduate Ambassador
- **Yale-NUS Alumni Mentorship Program** *2022*
  Alumni Mentor

Awards and Honors

- Harvard Quantum Initiative Prize Postdoctoral Fellowship *2023*
- Yale Doctoral Thesis with Distinction *2023*
- Shanghai Outstanding Graduate (top 5%) *2017*
- Honor Student (Physics) in National Top Undergraduate Training Program *2017*
- Xueersi Fellowship (top 3%) *2016*

Mentoring

U=Undergraduate, G=Graduate

[U.4] Felipe Abreu, *MIT*, **Magnetometry of a levitated magnet** *2026–present*

[G.10] Addison NewRingeisen, *Harvard*, **Driving mechanical resonators via a single electron** *2024–present*

[G.9] Alexander Polo, *Harvard*, **Exploring dynamics of vortices in superconductors** *2024*

[G.8] Binhan Hua, *Harvard*, **Toward entangling distant NVs via mechanical resonators** *2023–2025*

[G.7] Trisha Madhavan, *Harvard*, **Levitated magnets: sensing and quantum coupling** *2023–present*

[G.6] Frankie Fung, *Harvard*, **Toward a mechanical interface for mediating spin-spin interactions** *2023–2025*

[G.5] DaLi Schaefer, *Harvard*, **Toward high-cooperativity spin-mechanics with levitated magnets** *2023–2024*

[G.4] Theophilus Human, *Yale*, **Levitated superfluid helium drops for optomechanics** *2022–2023*

[G.3] Igor Brandao, *Yale*, **Levitated superfluid helium drops for optomechanics** *2022–2023*

[G.2] Daniel Sibilia, *Yale*, **Magnetic levitation of helium-3** *2022*

[U.3] Juan Recaoro, *Yale*, **Optomechanics in supercooled water droplets** *2022–2023*

[U.2] Giovanna Turong, *Yale*, **Magnetically levitated supercooled water droplets** *2021–2023*

[G.1] Chitres Guria, *Yale*, **Levitated superfluid helium drops for optomechanics** *2019*

[U.1] Mehmet Tuna Uysal, *Yale*, **Optomechanics of whispering gallery modes in liquid helium** *2018–2019*

Teaching

- **PHYSICS 271: Topics in the Physics of Quantum Information** *Fall 2024*
  External teaching assistant (final presentations) for M. D. Lukin Harvard
- **PHYS 201: Fundamentals of Physics** *Spring 2020*
  Graduate teaching assistant for D. Poland Yale
- **PHYS 508: Quantum Mechanics I** *Fall 2019*
  Graduate teaching assistant for W. Goldberger Yale
- **PHYS 4710: Independent Projects in Physics** *2019–2023*
  Undergraduate independent research projects mentor for J. G. E. Harris Yale

References

1. **Mikhail D. Lukin**
   Professor, Physics Department
   Harvard University
   Email: lukin@physics.harvard.edu
   Phone: +1(617) 495-2862
   Relationship: *Postdoc Advisor*

2. **Jack G. E. Harris**
   Professor, Physics Department
   Yale University
   Email: jack.harris@yale.edu
   Phone: +1(203) 432-3826
   Relationship: *Ph. D. Advisor*

3. **Benjamin A. Stickler**
   Professor, Institute for Complex Quantum Systems
   Ulm University
   Email: benjamin.stickler@uni-ulm.de
   Phone: +49(731) 50-22820
   Relationship: *Collaborator*

4. **Simon Gröblacher**
   Professor, Kavli Institute of Nanoscience Delft
   Delft University of Technology
   Email: s.groeblacher@tudelft.nl
   Phone: +31(15) 278-6124
   Relationship: *Collaborator*